PFD473S – Professional Development/Methods

EXAM STUDY GUIDE | AON & PERT | 2026

Based on April 2023 & March 2024 Past Papers

Explained Simply — You've Got This! 💪

SECTION 1: AON Node — The Building Block

Think of an AON (Activity on Node) diagram like a **LEGO map** for your project. Each activity is a LEGO brick. The bricks connect with arrows to show what must happen first.

What Does One Node Look Like?

Every box (node) holds SIX pieces of information:

Here is a sample AON node for Activity A (duration = 4 days):

Position in Box	Name	Meaning (Simple)	Formula
TOP LEFT	ES – Early Start	The EARLIEST day the activity CAN start	ES = largest EF of all predecessors
TOP RIGHT	EF – Early Finish	The EARLIEST day the activity CAN finish	EF = ES + Duration
MIDDLE	Activity Name	The name / letter of the task	—
3rd row	Duration (d)	How many days the task takes	Given in question
BOTTOM LEFT	LS – Late Start	The LATEST day it CAN start WITHOUT delaying project	LS = LF − Duration
BOTTOM RIGHT	LF – Late Finish	The LATEST day it CAN finish WITHOUT delaying project	LF = smallest LS of all successors

Sample node — Activity A, duration 4, ES=0, EF=4, LS=0, LF=4:

ES = 0	EF = 4
Activity A	
Duration = 4	
LS = 0	LF = 4

💡 **REMEMBER:** *RED/shaded nodes = CRITICAL PATH. If these activities are late, the WHOLE project is late!*

SECTION 2: Forward Pass (Left to Right)

Imagine pouring water from a tap. It flows forward through the network. The Forward Pass tells us the EARLIEST times.

Two Simple Rules:

- Rule 1: If an activity has NO predecessor → ES = 0 (it starts on Day 0)
- Rule 2: If an activity HAS predecessors → ES = the BIGGEST EF among all predecessors (take the MAXIMUM)
- EF always = ES + Duration (this never changes!)

FORWARD PASS FORMULAS:

ES = MAX (EF of all predecessors) [or 0 if no predecessors] **EF = ES + Duration**

Why MAX? (Think of it like this...)

Imagine you are baking a cake 🎂. Before you can decorate it you need BOTH the cake to be baked AND the icing to be made. If the cake takes 2 hours and the icing takes 5 hours, you cannot start decorating until BOTH are done — so you must wait 5 hours (the biggest one). That is why we take the MAX!

SECTION 3: Backward Pass (Right to Left)

Now pour water backwards! The Backward Pass tells us the LATEST times. We start from the END of the project and work backwards.

Two Simple Rules:

- Rule 1: The LAST activity's LF = the project's total duration (from the Forward Pass)
- Rule 2: LF of an activity = the SMALLEST LS of all its successors (take the MINIMUM)
- LS always = LF − Duration (this never changes!)

BACKWARD PASS FORMULAS:

LF = MIN (LS of all successors) [or project duration if no successors] **LS = LF − Duration**

Why MIN? (Another cake example...)

After the cake is decorated, you need to deliver it to the party. The party starts at 5pm. You have two delivery drivers — Driver A can leave at 4:30pm, Driver B at 4:00pm. The latest you can START decorating is limited by the driver who needs to leave EARLIEST (4:00pm = the minimum). So we take the MIN.

SECTION 4: Critical Path & Float

Total Float (TF) — How Late Can This Activity Be?

Total Float is the number of days an activity can be DELAYED without delaying the whole project.

FLOAT FORMULAS:

Total Float (TF) = LS − ES (or = LF − EF, same answer!) **Free Float (FF) = ES of next activity − EF of this activity** **Independent Float (IF) = ES of next activity − LF of previous activity − Duration**

Critical Path Rule:

- Any activity with Total Float = 0 is on the CRITICAL PATH
- The Critical Path is the LONGEST path through the network
- It equals the total project duration

💡 **REMEMBER:** *If TF = 0, the activity is CRITICAL. One day late = whole project is one day late!*

The THREE types of float:

Float Type	Simple Meaning	Formula
Total Float (TF)	How long can I delay this WITHOUT delaying the whole project?	LS − ES
Free Float (FF)	How long can I delay this WITHOUT delaying the NEXT activity?	ES(next) − EF(this)
Independent Float (IF)	How long can I delay this even if ALL other activities took their maximum time?	ES(next) − LF(prev) − Duration

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SECTION 5: WORKED EXAMPLE — 2023 Paper Question 3

This is exactly the type of question you will get. Follow EVERY step!

Step 1 — The Activity Table (Given in Exam)

Activity	Duration (days)	Predecessor(s)
A	2	None (starts on Day 0)
B	3	A
C	4	None (starts on Day 0)
D	4	C
E	2	B
F	2	A and D (BOTH must finish first)
G	9	E and F (BOTH must finish first)

Step 2 — Draw the Network (AON)

Draw arrows FROM predecessor TO successor. Activities with no predecessor connect from START. The network looks like this:

Network Path	Description
Start → A → B → E → G → Finish	One path through the top
Start → A → F → G → Finish	A also feeds into F
Start → C → D → F → G → Finish	Another path from C

Step 3 — Forward Pass (Calculate ES and EF)

Start from the LEFT. Work right. Use MAX for activities with multiple predecessors.

Activity	Predecessors' EF Values	ES = MAX of above	Duration	EF = ES + Duration
A	None	0	2	0 + 2 = 2
C	None	0	4	0 + 4 = 4
B	A finishes at EF=2	2	3	2 + 3 = 5
D	C finishes at EF=4	4	4	4 + 4 = 8
E	B finishes at EF=5	5	2	5 + 2 = 7
F	A finishes EF=2 AND D finishes EF=8 → MAX(2,8)	8	2	8 + 2 = 10
G	E finishes EF=7 AND F finishes EF=10 → MAX(7,10)	10	9	10 + 9 = 19

☑ **Project Duration = EF of last activity = 19 DAYS**

Step 4 — Backward Pass (Calculate LF and LS)

Start from the RIGHT (Day 19). Work LEFT. Use MIN for activities with multiple successors.

Activity	Successors' LS Values	LF = MIN of above	Duration	LS = LF − Duration
G	Last activity → LF = 19 (project end)	19	9	19 − 9 = 10
F	G has LS = 10	10	2	10 − 2 = 8
E	G has LS = 10	10	2	10 − 2 = 8
D	F has LS = 8	8	4	8 − 4 = 4
B	E has LS = 8	8	3	8 − 3 = 5
C	D has LS = 4	4	4	4 − 4 = 0
A	B has LS=5 AND F has LS=8 → MIN(5,8)	5	2	5 − 2 = 3

Step 5 — Complete Node Summary Table

Activity	ES	EF	LS	LF	TF = LS−ES	Critical?
A	0	2	3	5	3	No
B	2	5	5	8	3	No
C	**0**	**4**	**0**	**4**	**0 ✓**	**YES ★**
D	**4**	**8**	**4**	**8**	**0 ✓**	**YES ★**
E	5	7	8	10	3	No
F	**8**	**10**	**8**	**10**	**0 ✓**	**YES ★**
G	**10**	**19**	**10**	**19**	**0 ✓**	**YES ★**

★ **CRITICAL PATH = C → D → F → G (Total = 4+4+2+9 = 19 days)**

💡 **REMEMBER:** *The critical path is the LONGEST path. Always add up the durations of the red activities to verify!*

SECTION 6: WORKED EXAMPLE — 2024 Paper Question 1 (with Floats)

This question also asks for Free Float and Independent Float for 3 non-critical activities.

Activity Table (Given)

Activity	Duration	Predecessor(s)
A	4	None
B	6	A
C	5	A
D	8	A
E	9	B
F	10	B
G	6	C and E
H	8	D
I	3	D
J	2	G and H

Network Structure

Path	Activities
Path 1 (CRITICAL)	A → B → E → G → J
Path 2	A → B → F (terminal — no successor)
Path 3	A → C → G → J
Path 4	A → D → H → J
Path 5	A → D → I (terminal — no successor)

Forward Pass Results

Activity	Predecessors' EF	ES	Duration	EF
A	None	0	4	4
B	A=4	4	6	10
C	A=4	4	5	9
D	A=4	4	8	12
E	B=10	10	9	19
F	B=10	10	10	20
G	C=9 and E=19 → MAX=19	19	6	25
H	D=12	12	8	20
I	D=12	12	3	15

Activity	Predecessors' EF	ES	Duration	EF
J	G=25 and H=20 → MAX=25	25	2	27

☑ **Project Duration = 27 DAYS** *(F finishes at 20 but J finishes at 27 — we take the LATEST end)*

Backward Pass Results

Activity	Successors' LS	LF	Duration	LS
J	Last activity → 27	27	2	25
G	J=25	25	6	19
H	J=25	25	8	17
F	No successor → 27 (project end)	27	10	17
E	G=19	19	9	10
I	No successor → 27	27	3	24
D	H=17 and I=24 → MIN=17	17	8	9
C	G=19	19	5	14
B	E=10 and F=17 → MIN=10	10	6	4
A	B=4, C=14, D=9 → MIN=4	4	4	0

Complete Summary Table

Activity	ES	EF	LS	LF	TF	Critical?
A	**0**	**4**	**0**	**4**	**0 ✓**	**YES ★**
B	**4**	**10**	**4**	**10**	**0 ✓**	**YES ★**
C	4	9	14	19	10	No
D	4	12	9	17	5	No
E	**10**	**19**	**10**	**19**	**0 ✓**	**YES ★**
F	10	20	17	27	7	No
G	**19**	**25**	**19**	**25**	**0 ✓**	**YES ★**
H	12	20	17	25	5	No
I	12	15	24	27	12	No
J	**25**	**27**	**25**	**27**	**0 ✓**	**YES ★**

CRITICAL PATH = A → B → E → G → J (4+6+9+6+2 = 27 days)

Float Calculations for 3 Non-Critical Activities

The question asks for Free Float, Independent Float, and Total Float for THREE non-critical activities. Use C, D, and H:

FLOAT FORMULAS (remember these!):

Total Float (TF) = LS − ES
Free Float (FF) = ES(successor) − EF(this activity)
Independent Float (IF) = ES(successor) − LF(predecessor) − Duration

Activity C (predecessor=A, successor=G):

- TF = LS − ES = 14 − 4 = 10 days
- FF = ES(G) − EF(C) = 19 − 9 = 10 days [C can be 10 days late without delaying G]
- IF = ES(G) − LF(A) − Duration(C) = 19 − 4 − 5 = 10 days

Activity D (predecessor=A, successors=H and I):

- TF = LS − ES = 9 − 4 = 5 days
- FF = MIN(ES of H, ES of I) − EF(D) = MIN(12,12) − 12 = 0 days [cannot delay without affecting H or I]
- IF = MIN(ES of H, ES of I) − LF(A) − Duration(D) = 12 − 4 − 8 = 0 days

Activity H (predecessor=D, successor=J):

- TF = LS − ES = 17 − 12 = 5 days
- FF = ES(J) − EF(H) = 25 − 20 = 5 days
- IF = ES(J) − LF(D) − Duration(H) = 25 − 17 − 8 = 0 days

💡 **REMEMBER:** *Independent Float is often 0 or negative (if negative, write 0). It is the most STRICT measure of how much delay is truly safe.*

SECTION 7: PERT Analysis — When We Are Not Sure of Durations

CPM (what we did above) assumes we KNOW exactly how long each activity takes. But real life is uncertain! PERT uses THREE time estimates per activity:

Symbol	Name	Meaning
a (or O)	Optimistic Time	If EVERYTHING goes perfectly — best case scenario
m (or M)	Most Likely Time	What usually happens — your best guess
b (or P)	Pessimistic Time	If EVERYTHING goes wrong — worst case scenario

PERT FORMULAS (MEMORISE THESE!):

Expected Time: te = (a + 4m + b) ÷ 6
Variance: Var = [(b − a) ÷ 6]²
Std Deviation: σ (sigma) = √(sum of variances on critical path)
Z-score: z = (Target date − Expected finish) ÷ σ
Target date: x = (z × σ) + Expected finish

Why ÷ 6? (Simple explanation)

We are doing a WEIGHTED AVERAGE. The most likely time (m) is multiplied by 4 because it is the most important. Then we divide by 6 (because we have 6 parts: 1 optimistic + 4 most likely + 1 pessimistic = 6). Simple!

Z-score and Probability Table (Learn these values)

The Z-score tells us how many standard deviations away from the mean our target is. Use this table:

Z value	Probability of finishing by that time
-2.00	2.3%
-1.65	5%
-1.00	15.9%
0.00	50%
0.47	≈ 68%
0.53	≈ 70%
1.00	84.1%
1.28	90%
1.645 (use this for 95%)	95%
2.00	97.7%

💡 **REMEMBER:** *For '95% likelihood' questions, always use $z = 1.645$. Formula: $x = (1.645 \times \sigma) + \mu$*

SECTION 8: WORKED EXAMPLE — 2023 Paper Q4 (PERT)

Given in the exam:

Activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)	Predecessor
A	1	3	5	None
B	1	2	3	None
C	4	6	7	A and B
D	3	5	8	C
E	2	3	5	D
F	1	3	4	C
G	3	4	5	F
H	5	6	7	E
I	4	5	7	G and H

Step 1 — Calculate Expected Times (te)

Formula: $te = (a + 4m + b) \div 6$

Activity	Calculation	te (Expected Time)
A	$(1 + 4\times3 + 5) \div 6 = (1+12+5) \div 6 = 18\div6$	3.00
B	$(1 + 4\times2 + 3) \div 6 = (1+8+3) \div 6 = 12\div6$	2.00
C	$(4 + 4\times6 + 7) \div 6 = (4+24+7) \div 6 = 35\div6$	5.83
D	$(3 + 4\times5 + 8) \div 6 = (3+20+8) \div 6 = 31\div6$	5.17
E	$(2 + 4\times3 + 5) \div 6 = (2+12+5) \div 6 = 19\div6$	3.17
F	$(1 + 4\times3 + 4) \div 6 = (1+12+4) \div 6 = 17\div6$	2.83

Activity	Calculation	te (Expected Time)
G	(3 + 4×4 + 5) ÷ 6 = (3+16+5) ÷ 6 = 24÷6	4.00
H	(5 + 4×6 + 7) ÷ 6 = (5+24+7) ÷ 6 = 36÷6	6.00
I	(4 + 4×5 + 7) ÷ 6 = (4+20+7) ÷ 6 = 31÷6	5.17

Step 2 — Find All Paths and the Longest (Critical) Path

Note: Both A and B are predecessors of C, so they BOTH must finish before C can start. We find all paths:

Path	Activities	Total Duration
Path 1 ★ CRITICAL	A→C→D→E→H→I	3+5.83+5.17+3.17+6+5.17 = 28.34 days
Path 2	B→C→D→E→H→I	2+5.83+5.17+3.17+6+5.17 = 27.34 days
Path 3	A→C→F→G→I	3+5.83+2.83+4+5.17 = 20.83 days
Path 4	B→C→F→G→I	2+5.83+2.83+4+5.17 = 19.83 days

☑ **Expected Finish Time (μ) = 28.34 days** *(Path 1 is longest = Critical Path)*

Step 3 — Calculate Variances on the Critical Path

Formula: Var = [(b − a) ÷ 6]²   (only for activities ON the critical path!)

Activity (CP)	Calculation	Variance
A	[(5−1) ÷ 6]² = [4÷6]² = [0.667]²	0.444
C	[(7−4) ÷ 6]² = [3÷6]² = [0.50]²	0.250
D	[(8−3) ÷ 6]² = [5÷6]² = [0.833]²	0.694
E	[(5−2) ÷ 6]² = [3÷6]² = [0.50]²	0.250
H	[(7−5) ÷ 6]² = [2÷6]² = [0.333]²	0.111
I	[(7−4) ÷ 6]² = [3÷6]² = [0.50]²	0.250
	TOTAL VARIANCE = 0.444+0.250+0.694+0.250+0.111+0.250	2.00

σ (Standard Deviation) = √2.00 = 1.414

Step 4 — Q4.4: What is the probability of finishing by Day 29?

z = (Target − μ) ÷ σ = (29 − 28.34) ÷ 1.414 = 0.66 ÷ 1.414 = 0.47

z = 0.47   →   Probability ≈ 68%
Lookup z=0.47 in your Z-table → approximately 68% chance of finishing by Day 29

Step 5 — Q4.6: Time for 95% Likelihood of Finishing

For 95%, use z = 1.645. Rearrange to find x:

x = (z × σ) + μ = (1.645 × 1.414) + 28.34 = 2.33 + 28.34

x ≈ 30.67 days
There is a 95% chance the project will be done by Day 30.67 (≈ 31 days)

SECTION 9: WORKED EXAMPLE — 2024 Paper Q2 (PERT)

Activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)	Predecessor
A	4	5	7	None
B	1	3	6	A
C	4	5	7	A
D	3	7	10	B
E	4	6	9	B
F	2	3	5	C and D
G	1	7	10	E
H	2	6	8	E and F

Step 1 — Expected Times

Activity	Calculation	te
A	(4 + 4×5 + 7) ÷ 6 = 31÷6	5.17
B	(1 + 4×3 + 6) ÷ 6 = 19÷6	3.17
C	(4 + 4×5 + 7) ÷ 6 = 31÷6	5.17
D	(3 + 4×7 + 10) ÷ 6 = 41÷6	6.83
E	(4 + 4×6 + 9) ÷ 6 = 37÷6	6.17
F	(2 + 4×3 + 5) ÷ 6 = 19÷6	3.17
G	(1 + 4×7 + 10) ÷ 6 = 39÷6	6.50
H	(2 + 4×6 + 8) ÷ 6 = 34÷6	5.67

Step 2 — All Paths

Path	Activities & Durations	Total
Path 1 ★ CRITICAL	A(5.17)→B(3.17)→D(6.83)→F(3.17)→H(5.67)	24.01 days
Path 2	A(5.17)→B(3.17)→E(6.17)→G(6.50)	21.01 days
Path 3	A(5.17)→B(3.17)→E(6.17)→H(5.67)	20.18 days
Path 4	A(5.17)→C(5.17)→F(3.17)→H(5.67)	19.18 days

☑ **Expected Finish = 24.01 days (μ = 24.01)**

Step 3 — Variances on Critical Path (A→B→D→F→H)

Activity	Calculation	Variance
A	[(7−4)÷6]² = [0.50]²	0.250
B	[(6−1)÷6]² = [0.833]²	0.694

Activity	Calculation	Variance
D	$[(10-3)\div6]^2 = [1.167]^2$	1.361
F	$[(5-2)\div6]^2 = [0.50]^2$	0.250
H	$[(8-2)\div6]^2 = [1.00]^2$	1.000
	TOTAL VARIANCE	3.555

$\sigma = \sqrt{3.555} = 1.886$

Q2.3 — 95% Confidence Time

$x = (1.645 \times 1.886) + 24.01 = 3.10 + 24.01 = 27.11$ days

95% chance of finishing by approximately Day 27.1

Q2.4 — Probability of finishing 1 day AFTER expected (Day 25.01)

$z = (25.01 - 24.01) \div 1.886 = 1 \div 1.886 = 0.53$

$z = 0.53 \rightarrow$ Probability $\approx 70.2\%$
About 70% chance of finishing by 1 day after the expected finish time

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SECTION 10: Gantt Charts — The Visual Timeline

A Gantt chart is like a diary for your project. Each activity gets a row, and you shade in the days when that activity happens.

How to Draw a Gantt Chart — 4 Steps:

- Step 1: Work out when each activity STARTS (using its predecessor's finish date)
- Step 2: Work out when each activity ENDS (Start + Duration)
- Step 3: Draw a table — rows = activities, columns = days
- Step 4: Shade (mark X) the cells during which each activity runs

2024 Paper Q4 — Gantt Chart Example

Activities given:

Activity	Description	Predecessor	Duration (days)	Start Day	End Day
A	Train project team	None	10	Day 1	Day 10
B	Project paperwork & systems design	A	5	Day 11	Day 15
C	Modify purchased package	B	5	Day 16	Day 20
D	Manual systems flow	B	10	Day 16	Day 25
E	Modify in-house procedures	B	15	Day 16	Day 30

Activity	Description	Predecessor	Duration (days)	Start Day	End Day
F	Test & implement modifications (purchased)	C	15	Day 21	Day 35
G	Test & implement manual	D	15	Day 26	Day 40
H	Test & implement (in-house procedures)	E	5	Day 31	Day 35

☑ **Total Project Duration = 40 days (when Activity G finishes)**

Gantt Chart — Mark X in the appropriate blocks

In the exam you fill in X's on the chart. Here is the completed timing:

Activity	D1-5	D6-10	D11-15	D16-20	D21-25	D26-30	D31-35	D36-40
A – Train team	XXXXX	XXXXX						
B – Paperwork			XXXXX					
C – Modify pkg				XXXXX				
D – Manual flow				XXXXX	XXXXX			
E – In-house mods				XXXXX	XXXXX	XXXXX		
F – Test pkg mods					XXXXX	XXXXX	XXXXX	
G – Test manual						XXXXX	XXXXX	XXXXX
H – Test in-house							XXXXX	

💡 **REMEMBER:** *In the exam: each X block = 5 days. Count your predecessor's END day, then start the next activity the NEXT day.*

SECTION 11: Management Theory

11.1 — Project Management Definition (2024 Q3.1)

"Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements."
— PMBOK Guide

11.2 — The Fundamental Trade-off (2024 Q3.2)

The TRIPLE CONSTRAINT (Iron Triangle): Every project is constrained by THREE things. Changing one ALWAYS affects the others:

Constraint	Simple Meaning	Example of Trade-off
SCOPE (What you build)	The features and deliverables of the project — what you promised to deliver	If you ADD more features (bigger scope), you need MORE time OR MORE money

Constraint	Simple Meaning	Example of Trade-off
TIME (How long it takes)	The project deadline — when it must be done	If you SHORTEN the deadline, you need MORE people (costs more) OR deliver LESS
COST (What you spend)	The budget — how much money you have	If you CUT the budget, you must take MORE time OR deliver LESS

The trade-off: You can have it FAST, CHEAP, or GOOD — pick TWO. You cannot have all three!

11.3 — Henri Fayol's Principles (2024 Q3.3)

Fayol had 14 principles. The exam asks you to name FOUR and explain how they apply to a project manager with a team of 5 developers. Here are the best ones to know:

Principle	Simple Meaning	PM Example with 5-person team
1. Division of Work	Give each person a SPECIFIC task to specialise in. Don't have everyone doing everything.	The PM assigns: Developer 1 = database, Developer 2 = UI, Developer 3 = testing, etc. Each becomes expert in their area → faster and better quality.
2. Authority & Responsibility	Whoever has the POWER to give orders must also ACCEPT responsibility for results.	The PM has the authority to assign tasks to the 5 developers. But the PM is also responsible if the project fails — you cannot have power without accountability.
3. Unity of Command	Each employee should receive orders from only ONE boss.	Each of the 5 developers reports to only the PM — not to multiple managers. This avoids confusion about who to listen to.
4. Unity of Direction	Everyone works toward ONE common goal under ONE plan.	The whole team follows ONE project plan. Nobody goes off doing their own thing. All 5 developers aim for the same deadline and deliverables.
5. Subordination of Individual Interest	The PROJECT goal comes before personal goals.	If a developer wants to learn a new technology, that personal desire must not slow down the team's deadline. The project comes first.
6. Esprit de Corps (Team Spirit)	Build harmony and teamwork among staff.	The PM holds team meetings, celebrates wins together, and resolves conflicts quickly. A motivated team delivers better software.
7. Order	The right person must be in the right job at the right time.	The PM ensures the most skilled developer is assigned to the hardest task. Resources (tools, licenses, computers) are ready when needed.
8. Remuneration	Fair pay motivates workers.	Developers who go beyond expectations should be rewarded — through bonuses, recognition, or promotion. Unfair pay causes people to leave.

💡 **REMEMBER:** *In the exam: name the principle, define it in ONE sentence, then give a specific example using the project manager + 5-person team scenario.*

11.4 — Mintzberg's 10 Managerial Roles (2023 Q1)

Mintzberg grouped 10 roles into 3 categories. A PM plays ALL of these roles:

Category	Role	What the PM does
INTERPERSONAL (People Roles)	1. Figurehead	Represents the team at meetings, conferences, ceremonies. Signs off documents officially.
INTERPERSONAL	2. Leader	Motivates the team, sets direction, hires/fires, ensures everyone works efficiently.
INTERPERSONAL	3. Liaison	Communicates with external stakeholders, clients, vendors outside the team.
INFORMATIONAL (Info Roles)	4. Monitor	Scans the environment, tracks project progress, watches for risks and issues.
INFORMATIONAL	5. Disseminator	Shares important information WITH the team — project updates, decisions, changes.
INFORMATIONAL	6. Spokesperson	Speaks FOR the team to outsiders — reports to management, clients, shareholders.
DECISIONAL (Decision Roles)	7. Entrepreneur	Innovates — finds new solutions, improves processes, drives positive change.
DECISIONAL	8. Disturbance Handler	Resolves conflicts, technical problems, unexpected crises. Puts out fires!
DECISIONAL	9. Resource Allocator	Decides who gets what: assigns tasks, manages budget, allocates tools and time.
DECISIONAL	10. Negotiator	Negotiates contracts with clients, vendors, and stakeholders. Gets the best deal.

💡 **REMEMBER:** *Remember the 3 categories: Interpersonal (3 roles) → Informational (3 roles) → Decisional (4 roles). Total = 10 roles.*

SECTION 12: Quick Reference Card — Everything at a Glance

ALL Formulas

Formula	What it calculates
EF = ES + Duration	Early Finish (forward pass)
ES = MAX(EF of all predecessors) or 0 if first	Early Start (forward pass)
LS = LF − Duration	Late Start (backward pass)
LF = MIN(LS of all successors) or project end if last	Late Finish (backward pass)
TF = LS − ES (also = LF − EF)	Total Float
FF = ES(next activity) − EF(this activity)	Free Float
IF = ES(next) − LF(previous) − Duration	Independent Float
$te = (a + 4m + b) \div 6$	PERT Expected Time
$Var = [(b - a) \div 6]^2$	PERT Variance
$\sigma = \sqrt{\text{(sum of variances on critical path)}}$	Standard Deviation of project
$z = (\text{Target date} - \mu) \div \sigma$	Z-score (probability lookup)
$x = (z \times \sigma) + \mu$	Find date for given probability

Formula	What it calculates
For 95% confidence: use $z = 1.645$	95% probability target date

Critical Path Checklist — Do this in ORDER:

- ☐ Step 1: List all activities, durations, predecessors
- ☐ Step 2: Do FORWARD PASS → find ES and EF for every activity
- ☐ Step 3: Project duration = EF of the last activity
- ☐ Step 4: Do BACKWARD PASS → find LF and LS for every activity
- ☐ Step 5: Calculate TF = LS − ES for every activity
- ☐ Step 6: Activities with TF = 0 are on the CRITICAL PATH
- ☐ Step 7: Write the critical path as a sequence: A → B → E → G → J

PERT Checklist:

- ☐ Step 1: Calculate te for EVERY activity using $te = (a+4m+b)\div 6$
- ☐ Step 2: Find ALL paths and add up te values
- ☐ Step 3: The LONGEST path = Critical Path = Expected Finish (μ)
- ☐ Step 4: Calculate $Var = [(b-a)\div 6]^2$ for ONLY critical path activities
- ☐ Step 5: Total Variance = sum all CP variances; $\sigma = \sqrt{(\text{total variance})}$
- ☐ Step 6: For probability → find $z = (\text{target} - \mu) \div \sigma$ → look up z-table
- ☐ Step 7: For 95% → $x = (1.645 \times \sigma) + \mu$

YOU'VE GOT THIS! GOOD LUCK! 🎓